

MA 526: An Introduction to Wavelets

Course Code	MA 526
Course Name	An Introduction to Wavelets
L-T-P-C	3-0-0-3
Prerequisites	Basic Knowledge of Real Analysis and Linear Algebra
Intended for	UG/PG
Mutual Exclusion	None
Approval	46th BoA (May 2022)

Course Contents

- **Review of Linear Algebra:** Complex Series, Euler's Formula, Roots of Unity, Linear Transformations and Matrices, Change of Basis, diagonalization of Linear Transformations and Matrices, Inner Product, Orthogonal Bases, Unitary Matrices. (5 hours)
- **The Discrete Fourier Transform:** Definition and Basic Properties of Discrete Fourier Transform, Translation- Invariant Linear Transformations, The Fast Fourier Transform. (7 Hours)
- **Wavelets on Finite Group $\mathbb{Z}_N$:** Convolution, Fourier Transform on $\mathbb{Z}_N$, Definition of Wavelets and Basic Properties, Construction of Wavelets on $\mathbb{Z}_N$. (6 Hours)
- **Wavelets on Infinite Discrete Group $\mathbb{Z}$:** Definition and Basic Properties of Hilbert spaces, Complete orthonormal sets in Hilbert Spaces, The spaces $l_2(\mathbb{Z})$ and $L_2([- \rho, \rho])$, Basic Fourier Series, The Fourier Transform and Convolution on $L_2(\mathbb{Z})$ Wavelets on $\mathbb{Z}$. (8 Hours)
- **Wavelets on $\mathbb{R}$:** Convolution and Approximate Identities, Fourier Transform on $\mathbb{R}$, Bases for The Space $L_2(\mathbb{R})$, Belian-Low Theorem, Wavelets on $\mathbb{R}$, Multiresolution Analysis, Construction of Wavelets from multiresolution Analysis, Construction of Compactly supported Wavelets, Haar Wavelets, Band-Limited Wavelets, Applications. (16 Hours)

Laboratory Modules

Nil

Text Books

1. Michael W. Frazier, **An Introduction to Wavelets Through Linear Algebra**, Springer-Verlag, 1999.
2. Eugenio Hernandez, Guido Weiss, **A First Course on Wavelets**, CRC Press, 1996.

References

1. Ingrid Daubechies, **Ten Lectures on Wavelets**, CBMS -NSF Regional Conference Series in Applied Mathematics, 61. Society for Indus trial and Applied Mathematics (SIAM), 1992.
2. George Bachman, Lawrence Narici, Edward Beckenstein, **Fourier and Wavelet Analysis**, Springer-Verlag, 2000.
3. Howard L. Resnikoff, Raymond O. Wells, Jr., **Wavelet Analysis**, Springer-Verlag, 1998.